Supplementary Material: Characteristics and consequences of H_2O undersaturation and low melt fertility in mountain belt interiors

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1 Methods

Thin sections were made perpendicular to foliation and parallel to lineation where relevant. Major element mineral analyses were acquired using a Cameca SX-100 electron probe micro-analyser (EPMA) and a JEOL JXA-iHP200F Field Emission EPMA, both at the University of Cambridge. The accelerating voltage for all sessions was 20 kV and beam current 20 nA, with a beam size of 1 μm for silicates and 5 μm for feldspars, with 10 to 60 ms dwell times. Natural standards were used for calibration, and a ZAF correction procedure was used. Mineral chemistry data are provided in the Supplement.

Biotite formulae were calculated using the method of Li et al. (2020). Amphibole formulae were calculated using a spreadsheet provided by Prof. David Waters (personal correspondence), consistent with the Holland et al. (1994) scheme for Fe^{3+} calculation (modified by Dale et al., 2000), the Diener et al. (2012) a–X model for amphibole, and the recommended amphibole recalculation procedure of Hawthorne et al. (2012). All other mineral formulae were calculated using the program AX (version Oct-2019; Holland (2009)). Anorthite content, $X_{An} = Ca/(Ca + Na + K)$, is reported assuming ternary feldspar compositions to be consistent with the employed Holland et al., 2003 a–X model for feldspar. Mineral abbreviations follow the guidelines of Whitney et al. (2010).

Mineral modes were obtained via phase mapping using a FEI Quanta 650F Scanning Electron Microscope at the University of Cambridge. Two different programmes were used owing to the former having been decommissioned during the study: (1) Quantitative Evaluation of Minerals by Scanning Electron Microscopy (QEMSCAN), associated with twin energy dispersive spectrometers by Bruker, and (2) AZtecMineral, associated with a single Ultim Max energy dispersive spectrometer by Oxford Instruments. Mineral modes determined from thin section are assumed to be representative in the third-dimension so are quoted as vol.% below.

Model bulk rock compositions were calculated using measured mineral modes combined with representative mineral compositions (e.g. Palin et al., 2016). Phases not included in

the model system are primarily accessories (e.g. apatite and sulfides) and were therefore discounted. Measured modes were then renormalised. Garnet and clinopyroxene were idealised to conform to the stoichiometric criteria of their α–X models, e.g. tetrahedral Si in garnet was set as 3 cpfu. Bulk compositions are listed in Table 3.

2 Mineral chemistry

Table 1: Mineral compositional data for samples GT17, GT10-1, GT10-2, GT21, GT36-WR, GT45, GT60-M, GT58 and GT33. Data are presented as raw microprobe oxide analyses (wt.%), with cation totals calculated using AX Holland, 2009 and are representative single point analyses. Biotite formulae are calculated following the procedure of Li et al. (2020) and are charge balanced with $10(O^{2-}) + 2(Cl^{-}, F^{-}, OH^{-}, O^{2-}) = 12$. As F and Cl were not analysed in biotite, the oxygen total is 12. Intra-grain locations are specified for compositionally zoned minerals. $X_{Mg} = Mg/(Mg + Fe^{2+})$; $X_{Fe^{3+}} = Fe^{3+}/(Fe^{2+} + Fe^{3+})$; $X_{Alm} = Fe^{2+}/(Fe^{2+} + Mg + Ca + Mn)$; $X_{Grs} = Ca/(Fe^{2+} + Mg + Ca + Mn)$; $X_{Prp} = Mg/(Fe^{2+} + Mg + Ca + Mn)$; $X_{Sps} = Mn/(Fe^{2+} + Mg + Ca + Mn)$; $X_{An} = Ca/(Na + Ca + K)$. End-member proportions for clinopyroxene (diopside) assume a Wo-En-Fs-Jd-Ae solid solution such that $X_{Jd} = (Na + Al^{VI})/(Na + Fe_{total} + Mg + Ca + Al^{VI})$

Sample GT17

Mineral	Grt	Pl	Pl	Kfs	Bt	Ep
Location	Inner rim	Rim	Core	Inner rim	-	Mantle
SiO_2	36.75	67.11	63.79	63.99	39.46	37.74
TiO_2	0.01	0.00	0.00	0.00	1.73	0.09
Al_2O_3	20.10	20.41	22.65	18.36	16.53	24.19
Cr_2O_3	0.00	0.00	0.00	0.00	0.01	0.01
FeO	29.06	0.05	0.08	0.01	27.78	10.79
MnO	5.81	0.00	0.00	0.00	0.38	0.28
MgO	0.16	0.00	0.00	0.00	2.19	0.00
CaO	7.91	1.17	3.13	0.00	0.11	23.28
Na_2O	0.08	11.32	9.94	0.53	1.86	0.00
K_2O	0.00	0.12	0.17	16.51	7.61	0.01
Total	99.88	100.18	99.76	99.40	97.66	96.39
Oxygens	12.00	8.00	8.00	8.00	11.00	12.50
Si	2.98	2.94	2.82	2.99	2.72	3.00
Ti	0.00	0.00	0.00	0.00	0.14	0.01
Al	1.92	1.05	1.18	1.00	1.56	2.27
Fe^{3+}	0.12	0.00	0.00	0.00	0.39	0.71
Fe^{2+}	1.85	0.00	0.00	0.00	1.73	0.01
Mn	0.40	0.00	0.00	0.00	0.03	0.02
Mg	0.02	0.00	0.00	0.00	0.38	0.00
Ca	0.69	0.06	0.15	0.00	0.00	1.99
Na	0.01	0.96	0.85	0.04	0.02	0.00
K	0.00	0.01	0.01	1.00	0.95	0.00
Sum	8.00	5.02	5.02	5.03	7.91	8.00
X_{Mg}	0.01	-	-	-	0.18	0.00
$X_{Fe^{3+}}$	0.06	-	-	-	0.18	0.99
X_{Alm}	0.63	-	-	-	-	-
X_{Grs}	0.23	-	-	-	-	-
X_{Prp}	0.01	-	-	-	-	-
X_{Sps}	0.14	-	-	-	-	-
X_{An}	-	0.05	0.15	-	-	-
X_{Or}	-	-	-	0.97	-	-
X_{Ps}	-	-	-	-	-	0.24

Sample GT10

Mineral	Grt	Pl	Ms	Bt
Location	Inner rim	-	-	-
SiO_2	36.93	61.04	45.42	34.47
TiO_2	-	0.00	1.02	2.93
Al_2O_3	21.03	24.51	34.41	18.84
Cr_2O_3	-	0.00	0.00	-
FeO	33.76	0.01	1.33	20.16
MnO	2.48	0.00	0.01	0.06
MgO	3.62	0.00	0.73	9.41
CaO	2.15	5.99	0.00	-
Na_2O	0.02	7.85	0.81	0.14
K_2O	0.01	0.20	10.19	9.66
Total	100.00	99.60	93.92	95.67
Oxygens	12.00	8.00	11.00	12.00
Si	2.96	2.72	3.07	2.67
Ti	0.00	0.00	0.05	0.17
Al	1.99	1.29	2.74	1.72
Fe^{3+}	0.10	0.00	0.00	0.09
Fe^{2+}	2.17	0.00	0.08	1.22
Mn	0.17	0.00	0.00	0.00
Mg	0.43	0.00	0.07	1.09
Ca	0.19	0.29	0.00	0.00
Na	0.00	0.68	0.11	0.02
K	0.00	0.01	0.88	0.94
Sum	8.00	4.98	7.00	7.91
X_{Mg}	0.17	-	0.50	0.47
$X_{Fe^{3+}}$	0.04	-	0.00	0.07
X_{Alm}	0.73	-	-	-
X_{Grs}	0.06	-	-	-
X_{Prp}	0.15	-	-	-
X_{Sps}	0.06	-	-	-
X_{An}	-	0.30	-	-

Sample GT21

Mineral	Grt	Pl	Amp	Bt
Location	Inner rim	-	-	-
SiO_2	38.01	59.40	40.89	35.18
TiO_2	0.03	0.00	1.29	3.10
Al_2O_3	20.90	24.99	12.48	18.08
Cr_2O_3	0.02	0.00	0.00	0.05
FeO	25.62	0.12	18.88	19.04
MnO	2.20	0.00	0.29	0.03
MgO	3.70	0.00	8.64	9.05
CaO	10.37	7.10	11.62	0.01
Na_2O	0.01	7.04	1.24	0.31
K_2O	0.00	0.55	1.79	9.34
Total	100.86	99.20	97.12	94.19
Oxygens	12.00	8.00	23.00	12.00
Si	2.96	2.67	6.14	2.74
Ti	0.00	0.00	0.18	0.18
Al	1.92	1.33	2.29	1.66
Fe^{3+}	0.15	0.01	0.67	0.09
Fe^{2+}	1.52	0.00	1.81	1.15
Mn	0.15	0.00	0.05	0.00
Mg	0.43	0.00	1.91	1.06
Ca	0.87	0.34	1.86	0.00
Na	0.00	0.61	0.41	0.05
K	0.00	0.03	0.37	0.92
Sum	8.00	4.99	15.68	7.86
X_{Mg}	0.22	-	0.51	0.48
$X_{Fe^{3+}}$	0.09	-	0.27	0.07
X_{Alm}	0.74	-	-	-
X_{Grs}	0.08	-	-	-
X_{Prp}	0.16	-	-	-
X_{Sps}	0.01	-	-	-
X_{An}	-	0.38	-	-
Na_B [pfu]	-	-	0.07	-

Sample GT36

Mineral	Grt	Pl	Cpx	Amp	Bt
Location	Inner rim	Inner rim	Inner rim	Inner rim	-
SiO_2	37.70	62.51	50.61	41.43	34.96
TiO_2	0.06	0.00	0.19	2.40	5.68
Al_2O_3	20.72	22.92	2.86	11.17	13.70
Cr_2O_3	0.02	0.00	0.00	0.02	0.02
FeO	28.70	0.12	14.05	19.43	21.01
MnO	1.48	0.00	0.22	0.14	0.06
MgO	4.21	0.00	10.93	8.81	9.85
CaO	7.36	4.61	19.74	11.16	0.01
Na_2O	0.02	8.81	1.03	1.74	0.06
K_2O	0.00	0.64	0.01	1.58	9.64
Total	100.28	99.60	99.64	97.88	94.99
Oxygens	12.00	8.00	6.00	23.00	12.00
Si	2.97	2.79	1.92	6.31	2.78
Ti	0.00	0.00	0.01	0.27	0.32
Al	1.92	1.20	0.13	2.00	1.28
Fe^{3+}	0.14	0.00	0.10	0.50	0.21
Fe^{2+}	1.75	0.00	0.35	1.98	1.19
Mn	0.10	0.00	0.01	0.02	0.00
Mg	0.49	0.00	0.62	2.00	1.19
Ca	0.62	0.22	0.80	1.82	0.00
Na	0.00	0.76	0.08	0.51	0.01
K	0.00	0.04	0.00	0.31	0.97
Sum	8.00	5.01	4.00	15.72	7.95
X_{Mg}	0.22	-	0.64	0.50	0.50
$X_{Fe^{3+}}$	0.07	-	0.22	0.20	0.15
X_{Alm}	0.59	-	-	-	-
X_{Grs}	0.21	-	-	-	-
X_{Prp}	0.17	-	-	-	-
X_{Sps}	0.03	-	-	-	-
X_{An}	-	0.25	-	-	-
X_{Jd}	-	-	0.06	-	-
Na_B [pfu]	-	-	-	0.10	-

Sample GT45

Mineral	Grt	Pl	Amp	Bt
Location	-	Inner rim	-	-
SiO_2	36.57	61.36	40.93	33.97
TiO_2	0.07	0.00	1.89	4.48
Al_2O_3	20.13	24.13	12.19	15.60
Cr_2O_3	0.00	0.00	0.00	0.00
FeO	25.83	0.03	20.05	22.96
MnO	3.49	0.00	0.40	0.23
MgO	2.59	0.00	7.75	9.05
CaO	10.23	5.45	11.38	0.00
Na_2O	0.00	8.47	1.56	0.06
K_2O	0.01	0.26	1.89	9.67
Total	98.92	99.69	98.04	96.02
Oxygens	12.00	8.00	23.00	12.00
Si	2.93	2.73	6.25	2.69
Ti	0.00	0.00	0.22	0.25
Al	1.90	1.27	2.19	1.45
Fe^{3+}	0.22	0.00	0.45	0.23
Fe^{2+}	1.51	0.00	2.11	1.29
Mn	0.24	0.00	0.05	0.02
Mg	0.31	0.00	1.76	1.09
Ca	0.88	0.26	1.86	0.00
Na	0.00	0.73	0.46	0.01
K	0.00	0.02	0.37	0.96
Sum	8.00	5.01	15.73	7.99
X_{Mg}	0.17	-	0.46	0.46
$X_{Fe^{3+}}$	0.13	-	0.18	0.15
X_{Alm}	0.51	-	-	-
X_{Grs}	0.30	-	-	-
X_{Prp}	0.11	-	-	-
X_{Sps}	0.08	-	-	-
X_{An}	-	0.27	-	-
Na_B [pfu]	-	-	0.10	-

Sample GT60-M

Mineral	Grt	Pl	Kfs	Ms	Bt
Location	Inner rim	-	-	-	-
SiO_2	37.73	62.42	63.66	45.87	35.69
TiO_2	0.01	0.00	0.00	0.55	3.96
Al_2O_3	21.24	23.36	18.18	33.47	17.95
Cr_2O_3	0.00	0.00	0.00	0.01	0.00
FeO	30.60	0.03	0.25	1.82	18.70
MnO	1.39	0.00	0.00	0.00	0.03
MgO	3.88	0.00	0.00	0.82	9.38
CaO	5.89	4.94	0.07	0.01	0.01
Na_2O	0.02	8.99	1.26	0.31	0.08
K_2O	0.00	0.28	15.12	10.76	10.11
Total	100.76	100.02	98.54	93.62	95.91
Oxygens	12.00	8.00	8.00	11.00	12.00
Si	2.97	2.77	2.98	3.12	2.75
Ti	0.00	0.00	0.00	0.03	0.23
Al	1.97	1.22	1.00	2.68	1.63
Fe^{3+}	0.10	0.00	0.01	0.00	0.02
Fe^{2+}	1.92	0.00	0.00	0.10	1.20
Mn	0.09	0.00	0.00	0.00	0.00
Mg	0.46	0.00	0.00	0.08	1.08
Ca	0.50	0.24	0.00	0.00	0.00
Na	0.00	0.77	0.11	0.04	0.01
K	0.00	0.02	0.90	0.93	0.98
Sum	8.00	5.02	5.02	7.00	7.90
X_{Mg}	0.19	-	-	0.44	0.48
$X_{Fe^{3+}}$	0.05	-	-	0.00	0.02
X_{Alm}	0.65	-	-	-	-
X_{Grs}	0.17	-	-	-	-
X_{Prp}	0.15	-	-	-	-
X_{Sps}	0.03	-	-	-	-
X_{An}	-	0.25	-	-	-

Sample GT58

Mineral	Pl	Kfs	Amp	Bt	Ep
Location	-	-	-	-	-
SiO_2	61.58	64.45	40.47	35.99	37.73
TiO_2	0.00	0.00	1.04	2.98	0.13
Al_2O_3	23.61	18.28	12.00	14.74	23.12
Cr_2O_3	0.00	0.00	0.00	0.02	0.00
FeO	0.04	0.00	20.70	20.55	11.90
MnO	0.00	0.05	0.52	0.37	0.20
MgO	0.00	0.00	7.79	10.42	0.00
CaO	5.26	0.00	11.33	0.03	23.17
Na_2O	9.08	0.00	1.55	0.08	0.02
K_2O	0.17	0.79	1.87	10.09	0.00
Total	99.75	16.21	97.30	95.27	96.29
Oxygens	8.00	8.00	23.00	11.00	12.50
Si	2.76	2.99	6.22	2.82	3.01
Ti	0.00	0.00	0.12	0.17	0.01
Al	1.22	1.00	2.17	1.32	2.18
Fe^{3+}	0.00	0.00	0.64	0.25	0.79
Fe^{2+}	0.00	0.00	2.02	1.10	0.01
Mn	0.00	0.00	0.07	0.03	0.01
Mg	0.00	0.00	1.79	1.22	0.00
Ca	0.24	0.00	1.87	0.00	1.98
Na	0.77	0.07	0.46	0.01	0.00
K	0.01	0.96	0.37	0.98	0.00
Sum	5.02	5.02	15.73	7.90	8.00
X_{Mg}	-	-	0.47	0.53	-
$X_{Fe^{3+}}$	-	-	0.24	0.18	-
X_{An}	0.25	-	-	-	-
Na_B [pfu]	-	-	0.05	-	-
X_{Ps}	-	-	-	-	0.27

Sample GT33

Mineral	Grt	Pl	Bt
Location	-	-	-
SiO_2	38.48	63.81	36.94
TiO_2	0.00	0.00	1.51
Al_2O_3	20.85	21.97	16.48
Cr_2O_3	0.00	0.00	0.00
FeO	17.68	0.03	13.01
MnO	18.02	0.01	0.45
MgO	3.53	0.02	15.86
CaO	2.45	3.53	0.00
Na_2O	0.00	9.69	0.29
K_2O	0.02	0.12	9.55
Total	101.03	99.18	94.09
Oxygens	12.00	8.00	22.00
Si	3.04	2.84	2.77
Ti	0.00	0.00	0.09
Al	1.94	1.15	1.46
Fe^{3+}	0.00	0.00	0.20
Fe^{2+}	1.17	0.00	0.62
Mn	1.21	0.00	0.03
Mg	0.42	0.00	1.76
Ca	0.21	0.17	0.00
Na	0.00	0.84	0.04
K	0.00	0.01	0.92
Sum	7.99	5.01	7.89
X_{Mg}	0.26	-	0.74
$X_{Fe^{3+}}$	0.00	-	0.24
X_{Alm}	0.39	-	-
X_{Grs}	0.07	-	-
X_{Prp}	0.14	-	-
X_{Sps}	0.40	-	-
X_{An}	-	0.17	-

3 Sample chemistry

A summary of interpreted equilibrium assemblages is given in Table 2. The documented samples comprise four metasedimentary (psammite to pelite) and four meta-igneous (mafic to intermediate) compositions, and range from subsolidus (GT17, GT10, GT10-2, GT21, GT36-GC, GT36-A and GT33) to suprasolidus (GT60, GT45 and GT58) peak metamorphic conditions. The meta-igneous samples plot near the alkaline-sub-alkaline divide in total alkali-silica compositional space (Fig. 1), encompassing trachybasalt to trachyte compositions. Samples GT45 and GT21 plot in the same basaltic trachy-andesite field in Fig. 1 and are characterised by similar ferric iron ($X(\mathrm{Fe^{3+}}) = \frac{\mathrm{Fe^{3+}}}{\mathrm{Fe^{2+}}+\mathrm{Fe^{3+}}} = 0.28$ versus 0.29) and molar water content ($M(\mathrm{H_2O}) = 2.55$ versus 2.82), consistent with their similar mineral assemblages.

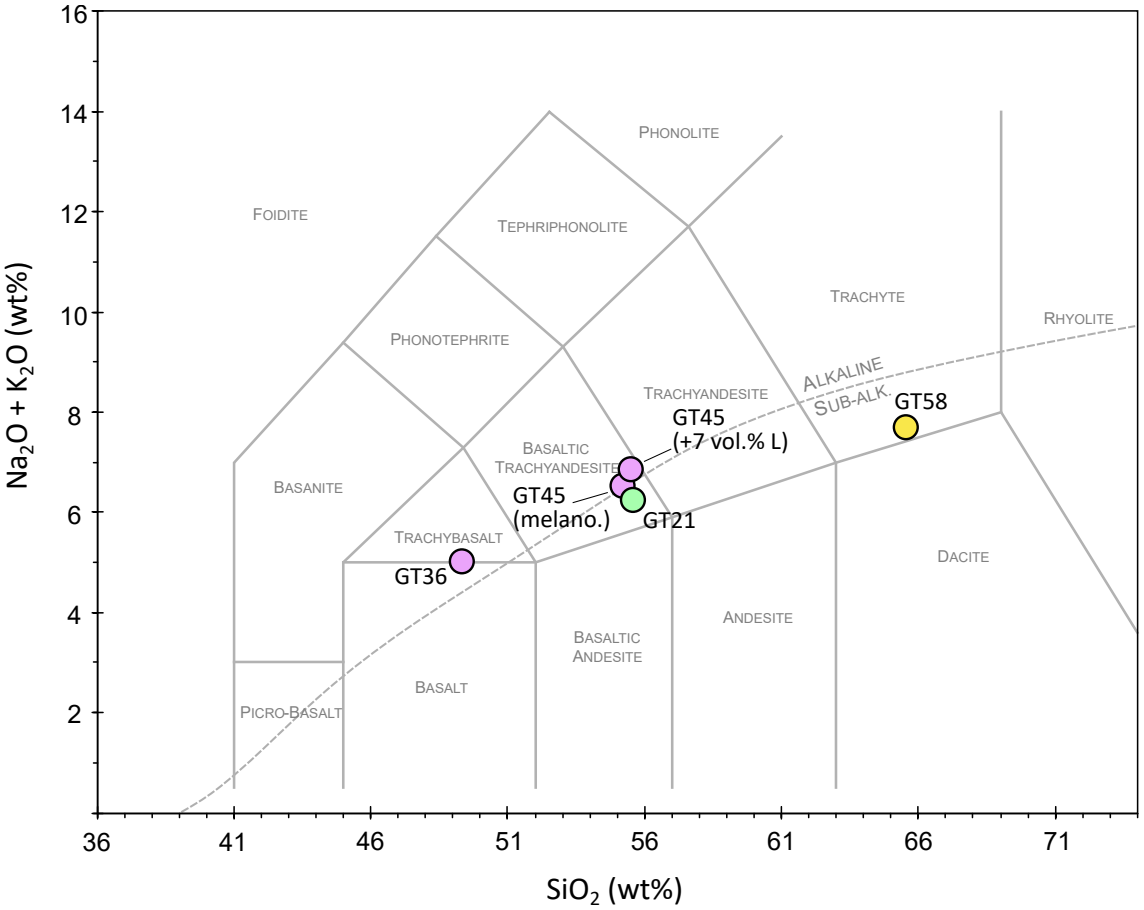


Figure 1: Study samples plotted on a total alkali–silica (TAS) diagram (Le Maitre et al., 2002). A single bulk composition is shown for sample GT36 based on an average of the Grt-Cpx and Amp domains (GT36-GC and GT36-A, respectively) as they feature similar bulk compositions. Sample bulk compositions are given in Table 3. Compositions are recalculated as anhydrous, with all Fe as FeO and normalised to 100 wt%.

Table 2: Petrographic summary of the eight study samples for which thermobarometry was undertaken. Distance measurements are relative to the Sole Thrust. The 'peak' column refers to the thermal peak assemblage identified in the thin section.

Sample	Dist (km)	Lithology	Peak	Retrograde
GT17	0.00	Grt-Ep psammite	Qz, Kfs, Olg, Bt, Ttn, Grt	Ep, Ab
GT10	1.11	Grt-Ky-Sil schist	Ms, Qz, Grt, Bt, Pl, Sil, Rt, Ilm	Ms, Chl
GT21	17.82	Grt amphibolite	Pl, Amp, Bt, Qz, Grt, Hem	Ilm, Bt
GT36-GC	32.74	Grt-Cpx amphibolite	Pl, Cpx, Grt, Bt, Ilm, Qz, Mag	Amp, Bt
GT36-A	32.74	Bt amphibolite	Amp, Pl, Cpx, Qz, Ilm, Bt, Kfs, Mag	Amp, Bt
GT45	63.76	Grt melanosome	Pl, Amp, Qz, Kfs, Bt, Mag, Grt, Ilm, L	Amp, Bt
GT60	84.81	Grt-Ky melanosome	Pl, Kfs, Ky, Grt, Bt, Rt, L	Qz, Ms, Sil
GT58	110.78	Ep-Bt amphibolite	Olg, Qz, Bt, L, Amp, Ttn	Bt, Ep, Ab
GT33	113.73	Grt-Chl schist	Pl, Bt, Qz, Ilm, Grt, Hem, Chl	Chl

Table 3: Bulk compositions used for phase equilibrium modelling, reported as moles. $X(Fe^{3+}) = Fe^{3+}/(Fe^{2+} + Fe^{3+})$. Samples GT36-A and GT36-GC correspond to the Amp domain and Grt-Cpx domain, respectively, whereas GT36 is the combined thin-section area. Sample GT60-M is the melanosome composition with reduced Fe^{3+} as determined in Supplementary Fig. S2. Bulk 1 and Bulk 2 are calculated melt-reintegrated compositions for melanosome samples GT60-M1.

Sample	Si	Al	Ca	Mg	Fe	K	Na	Ti	Mn	H	O	$X(Fe^{3+})$	$M(H_2O)$
GT17	79.69	14.77	0.92	0.50	3.07	8.27	4.75	0.18	0.05	2.64	194.64	0.25	1.32
GT10	61.26	26.74	0.79	3.48	7.66	7.49	1.66	0.65	0.36	15.44	188.64	0.04	7.72
GT21	57.80	22.93	6.97	5.40	8.17	3.13	7.81	0.73	-	5.65	181.46	0.29	2.82
GT36-A	50.83	18.16	8.70	7.72	11.44	1.84	8.37	2.64	-	6.87	171.62	0.19	3.43
GT36-GC	51.30	18.11	9.27	7.66	14.51	1.74	6.16	2.50	-	2.06	171.90	0.10	1.03
GT36	51.17	18.65	8.93	7.57	12.56	1.68	7.63	2.36	-	5.06	172.17	0.15	2.53
GT45	57.71	20.50	7.07	4.71	9.45	2.93	8.70	1.14	-	5.10	179.36	0.28	2.55
GT60-M	60.96	39.71	2.79	2.15	5.01	7.54	7.11	0.29	0.16	2.74	200.90	0.02	1.37
L1	38.50	22.66	0.49	0.04	0.13	7.79	11.56	0.00	0.00	79.68	161.15	0.02	39.8
L2	40.27	21.13	0.39	0.04	0.10	7.55	10.87	0.00	0.00	78.84	161.40	0.02	39.4
L3	40.84	20.64	0.35	0.04	0.10	7.36	10.82	0.00	0.00	78.53	161.48	0.02	39.3
Bulk 1	59.42	38.53	2.63	1.99	4.67	7.56	7.42	0.27	0.15	8.13	198.22	0.02	4.06
Bulk 2	58.08	37.32	2.48	1.86	4.35	7.56	7.66	0.25	0.14	13.08	195.65	0.02	6.54
GT58	68.90	20.29	3.92	3.17	3.83	4.06	9.48	0.60	-	4.45	189.79	0.23	2.22
GT33	53.36	20.68	1.99	8.44	9.11	4.38	9.64	3.28	0.65	8.79	177.32	0.30	4.39

4 Pseudosection analyses

4.1 $X(\mathbf{Fe^{3+}})$ constraints

The $X(Fe^{3+})$ content used in the modelling presented in the main text is based on charge balancing of idealised stoichiometry given by the minerals' a–X models. However, this

variable represents an unconstrained parameter as only total Fe was measured by EPMA. Owing to this limitation, accessory phases were predicted in some cases but not observed in thin section (samples GT17, GT36-GC, GT58 and GT33). Additional pseudosection analyses are given in Fig. S2 for these samples and demonstrate the sensitivity of interpreted thermal-peak assemblage fields to $X(\mathrm{Fe}^{3+})$.

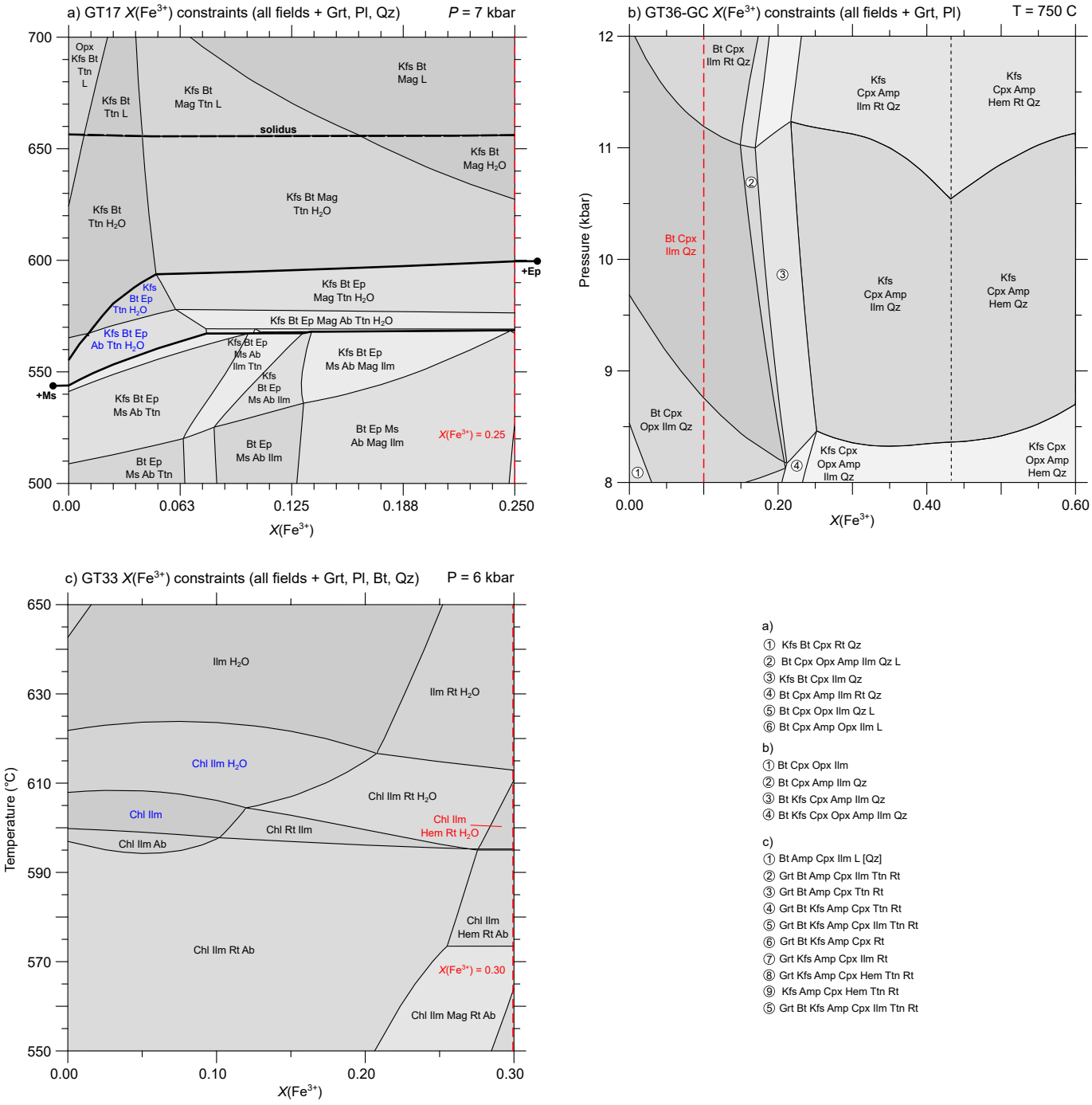


Figure 2: T–$X(\mathrm{Fe}^{3+})$ pseudosection results for samples GT17, GT36-GC, GT60-M and GT33. Stoichiometric $X(\mathrm{Fe}^{3+}$ is denoted by red dashed line and associated best-matching assemblage fields (as shown in the main text) by red text. Fields better matching the observed assemblages are denoted by blue text.

4.2 Sample GT60 – reintegration steps

Three melt reintegration steps were carried out for sample GT60, using melt compositions at 9.5 kbar in each case (L1 from GT60-M, L2 from Bulk 1, L3 from Bulk 2). Note that GT60-M refers to the melanosome portion of sample GT60. The first reintegrated composition was calculated as 0.93[GT60-M] + 0.07[L1], which was then used alongside 'L2' for the Bulk 2 calculation. These are shown in Fig. S3 below, with the initial GT60-M melanosome P–T pseudosection and final T–X_{melt} pseudosection shown in the main text (Fig. 11b).

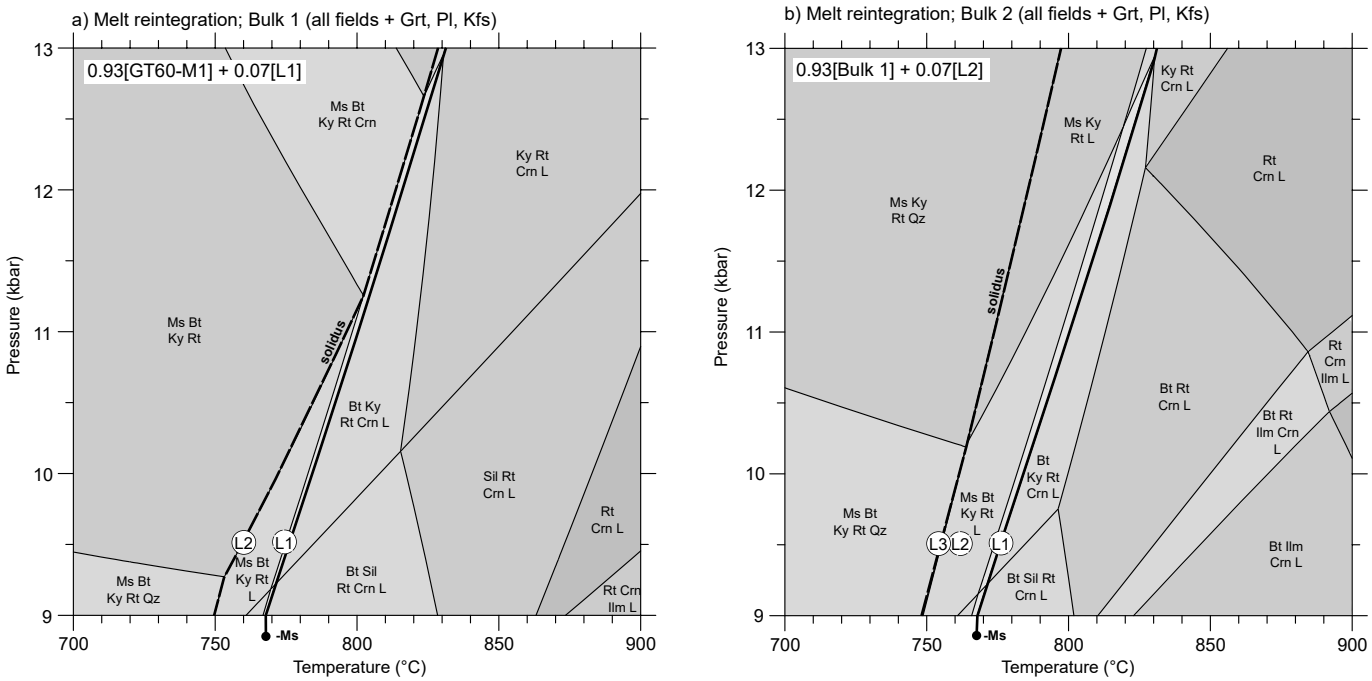


Figure 3: P–T pseudosection results for intermediate reintegration steps applied to melanosome sample GT60-M.

4.3 Sample GT17 – melt potential

A conceptual pseudosection at 10 kbar was produced for psammite sample GT17 to assess melt potential at H_2O saturation (Fig. 4. The 10 kbar constraint is used to be consistent with the same plots produced for other samples.

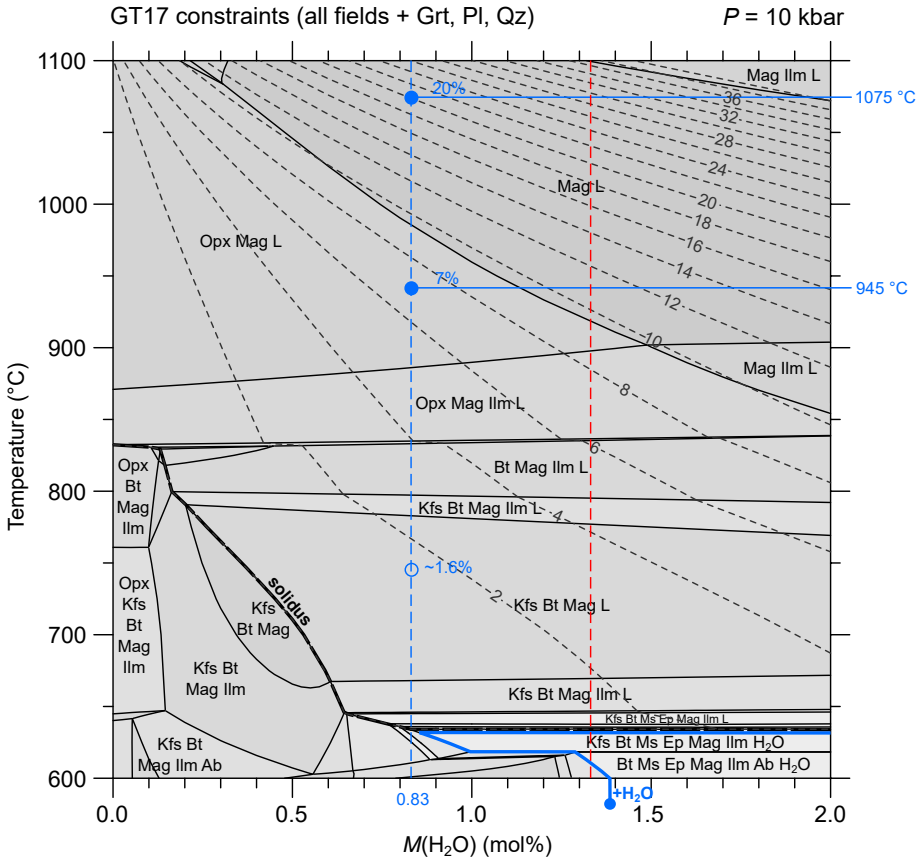


Figure 4: T–$M(H_2O)$ pseudosection produced for sample GT17 at 10 kbar, contoured for vol.% melt. Annotations are: the zero-mode isoline for H_2O (blue line), the $M(H_2O)$ for minimum saturation (dashed blue line), and temperatures for 7% and 20% melt (closed circles), and the stoichiometric $M(H_2O)$ (red dashed line).

5 Data location and projection method

The available P–T determinations from the Britt Domain and Shawanaga Domain are shown in Fig. S5. The data are located at a distance of up to ~60 km perpendicular to the study transect (i.e. the road, shown as a red line in Fig. S5). Thermobarometric data are based on calibrated reactions (e.g. 'GASP' and 'GRIPS' barometry, Grt-Bt thermometry) (Anovitz et al., 1990; Tuccillo et al., 1992) and the multi-equilibrium method 'TWQ' developed by Berman (1991) (Davidson et al., 1990; Ketchum et al., 1994; Jamieson et al., 1995; Bethune et al., 1997; Wodicka et al., 2000; Chapman, 2017).

P–T data were projected onto a profile (dashed line AB) to evaluate across-strike variations in these variables. The selected profile is parallel with the LITHOPROBE Line AB Long et al. (1996) and is 133 km in length. A distance value was measured for each data point, in kilometres relative to the Sole Thrust, along a series of lines parallel to the profile (dashed line AB), which contain each point. The distances were normalised to the total length of the lines, bounded by the Sole Thrust to the northwest and PSSZ to the southeast, and multiplied by the length of profile AB (130 km). The data were rescaled to the main profile from normalised distances instead of using absolute distances to account for along-strike variations in the widths of the Britt Domain and Shawanaga Domain. The across-strike extent of the Britt domain is unknown southwest of the present exposure of the Shawanaga

domain, where the ABT trends to the southeast and intersects the PSSZ. A value of 123 km, corresponding to the southern-most exposure of the PSSZ (Fig. S5), was chosen for data points up to ∼50 km southeast of the Sole Thrust whose 'lines' continue off-shore where the across-strike extent of the Britt domain is unknown. The resulting P–T profile is displayed in the main text in Fig. 16.

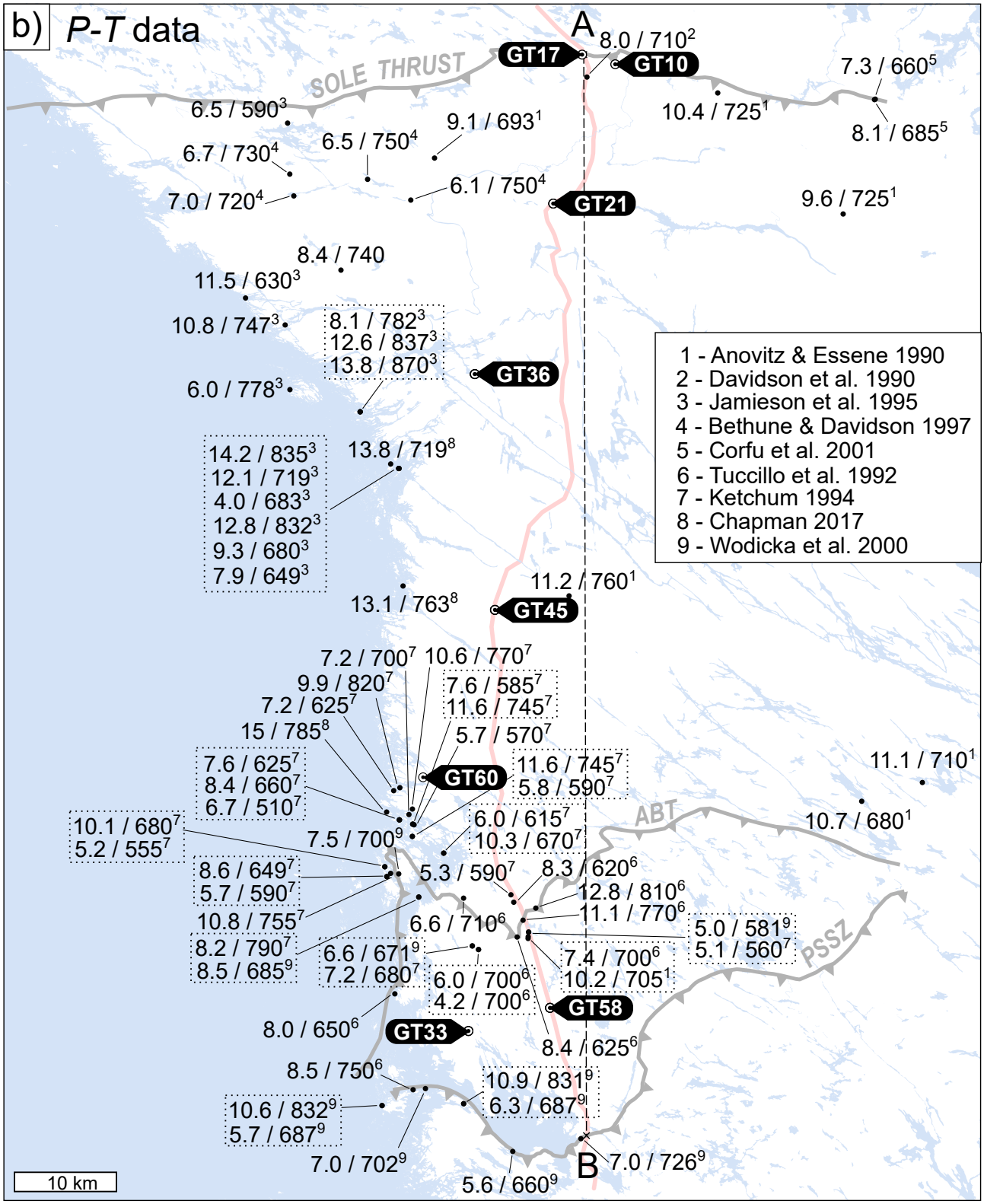


Figure 5: Location map of literature P (kbar) and T (°C) data from conventional thermobarometry and phase equilibrium modelling for the study area. The red line denotes the road along which study area samples were collected. Literature sources listed in the white box are indicated for each data point by a superscript. The locations of the samples discussed in the main text are shown by black tags.